Shanmukh Upadhyayula

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Education

University of Illinois Urbana-Champaign Urbana, IL
Bachelor of Science in Computer Science & Geographic Information Science *May 2027 (Expected)*

- GPA: 3.95/4.0
- Involvement: ACM Project Lead in Special Interest Groups for HCI and AI
- Relevant Coursework: Data Structures, Database Management in the Cloud, Algorithms & Models of Computation, Human-LLM Interaction, Deep Learning for Computer Vision, Virtual Reality, and Environmental Data Science

Technical Skills

Programming Languages: Python, Typescript, HTML/CSS, Javascript, R, C#, Java, C++, C, SQL
GIS Software: ArcGIS Pro, ArcGIS Online, ArcGIS Field Maps
Other Skills: React, Next.js, Streamlit, OpenCV, Geopandas, sf, Unity, PyTorch, XGBoost, PySal

Experience

Semantic Segmentation Failure Analysis - Python | **Paper** Urbana, IL
Undergraduate Researcher *May 2026 – August 2026*

- Cut segmentation error from 51% to under 10% at 20% coverage by engineering four label-free signals that rank a model's own errors without ground truth
- Proved failure detection survives domain shift (0.61-0.74 AUROC) across 4 benchmark datasets spanning 44 countries and 6 continents
- Diagnosed boundary failures at roughly half the accuracy of interior regions on a DeepLabV3-ResNet50 baseline, writing an IEEE-format paper and delivering a CyberGIS talk

Bias Reduction in Food Inspection Models across Chicago - Python | **Notebook** Urbana, IL
Undergraduate Researcher *May 2026 – August 2026*

- Cut Chicago's largest recall gap by 99%, to nearly zero, with a custom Group-DRO XGBoost objective, hand-derived gradient/Hessian, vs. per-group thresholds
- Traced the disparity to a ranking deficit, not a threshold artifact, via per-group ROC AUC well above chance over 8,312 inspections, correctly predicting which mitigation would work
- Rebuilt the City's forecasting pipeline on 2018-present data, cutting 148K raw records to 42K clean rows and clustering neighborhoods into 3 typologies via k-means and Ward clustering

Pathway to Improved Cities (ACM SIG-AIDA) - Python | **Repo** Urbana, IL
Project Lead *February 2026 – May 2026*

- Led a team of 8 building ensemble ML models (Random Forest, Gradient Boosting) with strong-fit R^2 for urban hardship, crime, crash risk, and transit access across all 77 community areas
- Developed an interactive Streamlit + Mapbox GL dashboard with choropleth maps, density heatmaps, LISA cluster visualizations, and a domain-aware upload system for on-demand ML prediction

Projects

MAS Sycophancy: Hierarchical Multi-Agent LLM Failure | *Python* | Paper March 2026 – May 2026

- Demonstrated that hierarchical agentic workflows collapse under a single injected authority hallucination, driving a large sycophancy effect and 100% cascade by turn 3
- Quantified the failure without an LLM judge across three independent signals (stance flips, planning errors, deference language), each absent in the uninjected baseline
- Showed that explicit anti-sycophancy prompting fails to prevent collapse, co-authoring an ACL-format paper on the resulting "Yes-Man" failure mode

Spatial Autocorrelation of AI Data Centers | *Python* | Notebook March 2026 – May 2026

- Proved 5 statistically significant hyperscale-impact effects vanish under a spatial-lag model, contrasting OLS against GM_Lag across 3,220 U.S. counties
- Surfaced a 34-county Northern Virginia electricity-price hot spot invisible to national statistics via bivariate Moran's I and LISA across 5 impact variables